

1. Right Angle Triangle Pattern

```
In [1]: for i in range(1, 6):
        print(' * ' * i)
```

```
*
* *
* * *
* * * *
* * * * *
```

2. Inverted Right Angle Triangle Pattern

```
In [2]: for i in range(5,0,-1):
        print(' * ' * i)
```

```
* * * * *
* * * *
* * *
* *
*
```

3. Pyramid Pattern

```
In [3]: for i in range (1,6):
        print('*(5-i)+' * '*(2*i-1))
```

```
*
* * *
* * * * *
* * * * * * *
* * * * * * * *
```

4. Inverted Pyramid Pattern

```
In [4]: for i in range(5,0,-1):
        print('*(5-i)+' * '*(2*i-1))
```

```
* * * * * * * *
* * * * * *
* * * *
* * *
*
```

5. Diamond Pattern

```
In [5]: for i in range(1,6):
        print('*(5-i)+' * '*(2*i-1))
        for i in range(4,0,-1):
            print('*(5-i)+' * '*(2*i-1))
```

```

*
* * *
* * * * *
* * * * * * *
* * * * * * * *
* * * * * *
* * * *
* * *
*

```

6. Hallow Square Pattern

```

In [6]: for i in range(5):
        for j in range(5):
            if i == 0 or i == 4 or j == 0 or j == 4:
                print('*',end='')
            else:
                print(' ',end='')
        print()

```

```

*****
*   *
*   *
*   *
*****

```

7. Full Square Pattern

```

In [7]: for i in range(5):
        print(' * '*5)

```

```

* * * * *
* * * * *
* * * * *
* * * * *
* * * * *

```

8. Right Angle Triangle (Number Pattern)

```

In [10]: for i in range(1, 6):
        print(' '.join(str(x) for x in range(1, i + 1)))

```

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

9. Inverted Right Angle Triangle (Number Pattern)

```

In [11]: for i in range(5,0,-1):
        print(' '.join(str(x) for x in range(1, i + 1)))

```

```

1 2 3 4 5
1 2 3 4
1 2 3
1 2
1

```

10. Floyd's Triangle

```

In [12]: num = 1
         for i in range(1,6):
             for j in range(1,i+1):
                 print(num, end=' ')
                 num+=1
             print()

```

```

1
2 3
4 5 6
7 8 9 10
11 12 13 14 15

```

11. Hallow Right Angle Triangle

```

In [14]: for i in range(1,6):
         for j in range(1,i+1):
             if j == 1 or j == i or i == 5:
                 print('*', end=' ')
             else:
                 print(' ', end=' ')
         print()

```

```

*
* *
*  *
*   *
* * * * *

```

12. Hallow Pyramid Pattern

```

In [15]: for i in range(1,6):
         for j in range(5 - i):
             print(' ', end=' ')
         for j in range(2 * i - 1):
             if j == 0 or j == 2 * i - 2 or i == 5:
                 print('*', end=' ')
             else:
                 print(' ', end=' ')
         print()

```

```

      *
     * *
    *  *
   *   *
  *    *
 *     *
* * * * *

```

13. Hallow Diamond Pattern

```
In [17]: n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print('*', end=' ')
        else:
            print(' ', end=' ')
    print()

for i in range(n - 1, 0, - 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print('*', end=' ')
        else:
            print(' ', end=' ')
    print()
```

```

      *
     * *
    *  *
   *   *
  *    *
 *     *
*      *
 *     *
  *    *
   *   *
    *  *
     * *
      *

```

14. Hallow Diamond (Number Pattern)

```
In [18]: n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()

for i in range(n - 1, 0, - 1):
    for j in range(n - i):
        print(' ', end=' ')
    for j in range(2 * i - 1):
        if j == 0 or j == 2 * i - 2:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()
```

```

      1
    2 2
  3   3
4    4
5    5
  4   4
    3 3
      2 2
        1

```

15. Butterfly Pattern

```

In [21]: n = 5
for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ', end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()
for i in range(n, 0, -1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ', end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()

n = 5
for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ', end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()
for i in range(n, 0, -1):
    for j in range(1, i + 1):
        print(j, end=' ')
    for j in range(2 * (n - i)):
        print(' ', end=' ')
    for j in range(1, i + 1):
        print(j, end=' ')
    print()

```

```

1                1
1 2              1 2
1 2 3            1 2 3
1 2 3 4          1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4          1 2 3 4
1 2 3            1 2 3
1 2              1 2
1                1
1                1
1 2              1 2
1 2 3            1 2 3
1 2 3 4          1 2 3 4
1 2 3 4 5 1 2 3 4 5
1 2 3 4 5 1 2 3 4 5
1 2 3 4          1 2 3 4
1 2 3            1 2 3
1 2              1 2
1                1

```

16. Hallow Number Pyramid

```

In [22]: n = 5
for i in range(1, n + 1):
    for j in range(n - i):
        print(' ', end=' ')

    for j in range(1, 2 * i):
        if j == 1 or j == 2 * i - 1 or i == n:
            print(i, end=' ')
        else:
            print(' ', end=' ')
    print()

```

```

      1
     2 2
    3   3
   4     4
  5 5 5 5 5 5 5

```

17. Full Star Pyramid

```

In [23]: n = 5

for i in range(1, n + 1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        print('*', end=' ')

    print()

```

```

      *
    * * *
  * * * * *
* * * * * * *

```

18. Inverted Full Star Pyramid

```

In [24]: n = 5

for i in range(n, 0, -1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(2 * i - 1):
        print('*', end=' ')

    print()

```

```

* * * * *
 * * * * *
  * * * *
   * * *
    *

```

19. Left Aligned Pyramid Pattern

```

In [25]: n = 5

for i in range(1, n + 1):

    for j in range(i):
        print('*', end=' ')
    print()

n = 5

for i in range(1, n + 1):
    for j in range(1, i + 1):
        print(j, end=' ')
    print()

```

```

*
* *
* * *
* * * *
* * * * *

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

20. Right Aligned Pyramid Pattern

```
In [26]: n = 5

for i in range(1, n + 1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(1, i + 1):
        print(j, end=' ')

    print()

n = 5

for i in range(1, n + 1):

    for j in range(n - i):
        print(' ', end=' ')

    for j in range(i):
        print('*', end=' ')

    print()
```

```
    1
  1 2
1 2 3
1 2 3 4
1 2 3 4 5
    *
  * *
* * *
* * * *
* * * * *
```